



**CraftAble: A Web-Based Ordering and Marketplace System for Filipino
Craftsmen for Competent Differently Abled Persons Inc., Utilizing Fuzzy Search
and Cosine Similarity Algorithms**

A Thesis Presented to the
Faculty of the College of Computing Studies
Eulogio "Amang" Rodriguez
Institute of Science and Technology

**In Partial Fulfillment of the Requirements
For the Degree of
Bachelor of Science in Information Technology**

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2026



CHAPTER 1

THE PROBLEM AND ITS BACKGROUND

This chapter includes the Introduction, Conceptual Framework, Statement of the problem, Scope and limitations, Significance of the study and the definition of terms used.

Background of the Study

The rapid growth of the digital economy in the Philippines, valued at over \$17 billion in 2025, has created new pathways for entrepreneurship and inclusive development. Yet, a significant "Digital Divide" persists for the 2.5 million registered Persons with Disabilities (PWDs) in the country (NCDA, 2025). While the Magna Carta for Disabled Persons (RA 7277) and the Philippine Digital Workforce Competitiveness Act (RA 11927) mandate equal participation, the technical architecture of mainstream e-commerce platforms remains fundamentally exclusionary. Local platforms such as Lazada, Shopee, and Zalora have transformed online shopping into a mainstream activity, but their mobile and web interfaces often prioritize profit and scalability over universal accessibility. The heavy use of JavaScript frameworks, poor semantic HTML structure, and the absence of ARIA (Accessible Rich Internet Applications) labels result in interfaces that are incompatible with screen readers and motor-assistive technologies. For example, visually impaired artisans struggle to navigate complex dashboards to upload product photos, while those with motor disabilities face significant friction in multi-step checkout systems that lack full keyboard or voice-command support. These barriers force PWD artisans to remain in physical marketplaces, limiting their reach and income in an increasingly contactless economy. This exclusion is not merely a technical inconvenience—it has serious social and economic consequences. PWD artisans are denied equal opportunities to participate in digital trade, reinforcing cycles of dependency and marginalization. The gap between policy and practice is evident: while government initiatives like the E-Commerce Philippines 2022 Roadmap emphasize inclusive growth, actual software implementations continue to prioritize scale



over the lived experience of disabled users. International studies, such as those by the World Bank (2020), confirm that inclusive digital platforms can directly improve livelihoods by enabling marginalized groups to participate in commerce. Thus, addressing accessibility is both a legal requirement and a social necessity for the Philippine IT landscape. This research proposes CraftAble An Inclusive Digital Marketplace, which moves beyond traditional retrofitting by adopting an Accessibility-First Development Framework. By integrating simplified hierarchical navigation, high-contrast UI modes, and screen-reader optimized management tools from the initial system architecture, CraftAble aligns with WCAG 2.1 Level AA standards. More importantly, it demonstrates how inclusive design can empower PWD artisans to achieve financial independence, provide buyers with transparent ways to support social enterprises, and offer policymakers a technical model of compliance with RA 7277 and RA 11927. In doing so, CraftAble aims to transform PWDs from passive beneficiaries into active economic contributors. Ultimately, this study situates PWD Artisans within both local and global movements toward inclusive technology. It aligns with the United Nations Sustainable Development Goals (SDGs), particularly SDG 8 (Decent Work and Economic Growth) and SDG 10 (Reduced Inequalities). By embedding accessibility into the very foundation of digital commerce, the project contributes to sustainable economic growth while ensuring that PWD artisans are recognized as empowered participants in the country's digital transformation.



Project Context

The development of CraftAble is situated within the current reality of PWD entrepreneurs in the Philippines who are struggling to keep up with the rapid shift toward a digital-first economy. At present, most PWD artisans rely on traditional methods of commerce, such as physical bazaars, local government trade fairs, or informal selling through social media platforms like Facebook and Messenger. While these channels provide some visibility, they are not designed with universal accessibility in mind. They often present high cognitive loads, cluttered interfaces, and a lack of structured business tools such as integrated inventory management or accessible checkout systems making it difficult for artisans with visual or motor impairments to manage a professional enterprise independently. Technically, the project is being developed to exist within a mobile-centric environment. Given that the majority of Filipino internet users access the web via smartphones, CraftAble is designed as a mobile-responsive website. This context requires a specific focus on ensuring the system architecture is compatible with native mobile assistive technologies, particularly TalkBack for Android and VoiceOver for iOS. By prioritizing semantic HTML and a clean, "Accessibility-First" UI, the study aims to ensure that the platform serves as a functional bridge to the digital market rather than another technical barrier. Furthermore, the project operates within a legal and social framework defined by the need for inclusive economic growth. Under RA 7277, PWDs are guaranteed equal access to social and economic participation, yet the digital divide remains a significant obstacle to financial independence. CraftAble addresses this by providing a specialized ecosystem that follows WCAG 2.1 standards, proving that e-commerce can be both a robust business tool and an inclusive space. Ultimately, this context ensures that the research is grounded in the real-world operational challenges faced by PWD sellers, aiming to transition them from the margins of physical trade into the center of the country's digital transformation.



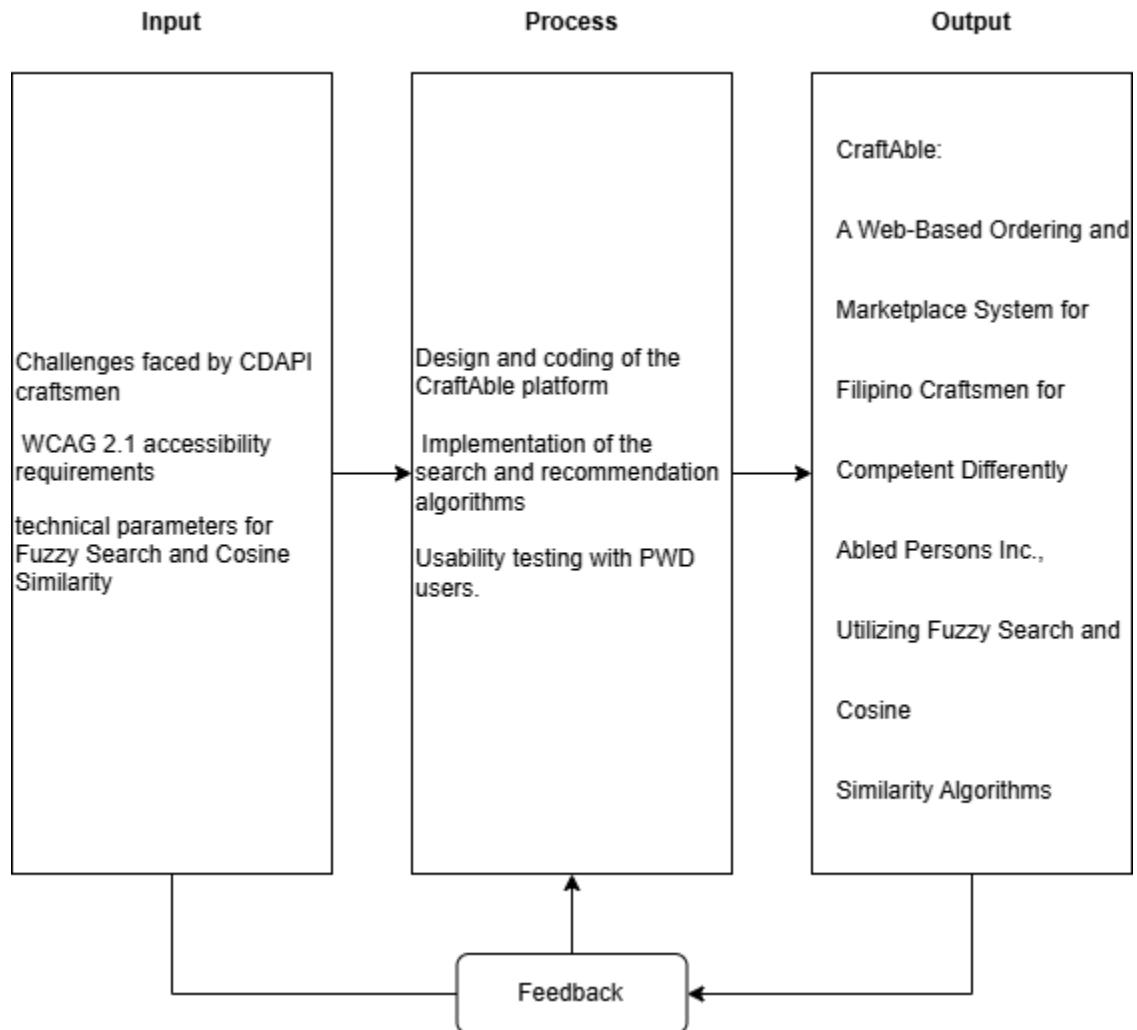
Purpose and Description

The primary purpose of CraftAble is to provide an accessible web-based ordering and marketplace system that empowers ACDAP craftsmen. The platform serves as a bridge between artisans and socially conscious consumers.

CraftAble is described as a high-accessibility web application built using the MERN Stack (MongoDB, Express.js, React, Node.js). To enhance the user experience, the system utilizes Fuzzy Search to account for typos or approximate spelling and Cosine Similarity Algorithms to provide accurate product recommendations based on artisan craftsmanship. These technical features ensure that products remain discoverable even with imprecise search terms, ultimately driving higher engagement and sales for the craftsmen.

Conceptual Framework

This research applies the "Input-Process-Output" or IPO model. The INPUT of the study determines the challenges faced by Filipino PWD artisans in accessing digital marketplaces, the requirements for WCAG 2.1 compliance, and the legal frameworks like RA 7277. The PROCESS involves the design and development of the PWD-Artisan platform, accessibility testing, and data analysis of system usability. The OUTPUT of the study would result in an accessible, inclusive, and transparent digital marketplace for PWD crafts





Objectives

To design, develop, and implement CraftAble, an inclusive web-based ordering and marketplace system for the Competent Differently Abled Persons Inc. (ACDAP), which integrates Fuzzy Search and Cosine Similarity Algorithms to enhance product discoverability and provide an accessible digital livelihood platform for Filipino PWD craftsmen.

Design and develop a web-based e-commerce interface using the MERN Stack (MongoDB, Express.js, React, Node.js) that strictly adheres to WCAG 2.1 standards to ensure full accessibility for ACDAP members with visual and motor impairments.

Implement Fuzzy Search logic within the marketplace search engine to improve the discoverability of PWD-made crafts by handling approximate string matches and potential typographical errors.

Integrate the Cosine Similarity Algorithm to generate accurate, content-based product recommendations for buyers, thereby increasing the market visibility and sales potential for PWD craftsmen.

Incorporate a Social Impact Dashboard that visualizes the direct correlation between consumer purchases and the livelihood milestones of ACDAP artisans.

Evaluate the technical effectiveness of the system based on the ISO 25010 software quality standards, specifically focusing on:

Functionality: Ensuring the ordering and marketplace features work as intended.

Usability: Assessing the ease of use for both the PWD artisans and the consumers.

Accessibility: Validating that the platform is perceivable and operable for users with disabilities.



Significance of the Study

the following are the target beneficiaries of this study:

ACDAP: Provides a specialized platform to sell crafts, bypass physical barriers, and gain financial autonomy.

Consumers: Offers an easy-to-use platform to support social enterprises with an intelligent search system that finds products even with vague queries.

The Field of Information Technology: Demonstrates how mathematical algorithms like Cosine Similarity can be applied to social-good projects to improve accessibility and inclusivity.

Scope and Limitations of the Study

The primary scope of this research is the development of CraftAble, a web-based marketplace for the Competent Differently Abled Persons Inc. (ACDAP). The system features an artisan portal for inventory management and a buyer interface with an intelligent search engine.

The study is delimited to ACDAP members within the Philippines. The technical scope focuses on the application of Fuzzy Search (for approximate string matching) and Cosine Similarity (for content-based filtering/recommendations).

The limitations include the reliance on stable internet connectivity for the artisans. Furthermore, while the system adheres to WCAG 2.1, it does not provide specialized hardware (like Braille displays), assuming users have their own assistive tools or basic digital literacy.



Definition of Terms

- **Accessibility.** Design ensuring PWDs can navigate and interact with the platform.
- **ACDAP.** Competent Differently Abled Persons Inc., the primary beneficiary and partner.
- **Cosine Similarity Algorithm.** A metric used to calculate product recommendations.
- **CraftAble.** The official name of the proposed web-based system.
- **Fuzzy Search.** A technique identifying results that match patterns approximately to handle typos.
- **ISO 25010.** International standard for software quality evaluation.
- **MERN Stack.** The technology stack (MongoDB, Express, React, Node) used for development.
- **RA 7277.** The Magna Carta for Disabled Persons, providing the legal basis for the study.
- **WCAG 2.1.** Technical standards for web accessibility.



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